

Calculus 2

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Limits and Continuity of Multivariate Functions

Problem 1

Determine the necessary and sufficient conditions on the coefficients a , b , and c such that the quadratic form $f(x, y) = ax^2 + bxy + cy^2$ is of the same order as $g(x, y) = x^2 + y^2$ as $(x, y) \rightarrow (0, 0)$.

Hint 1

To prove two functions are of the same order, you must show that the limit of their ratio is bounded between two positive constants m and M as the variables approach the origin.

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Hint 2

Utilize the transformation to polar coordinates: $x = r \cos \theta$ and $y = r \sin \theta$. Observe how the radius r cancels out in the ratio of the two quadratic forms.

Hint 3

Recall that a continuous function on a compact set (like $\theta \in [0, 2\pi]$) attains its maximum and minimum values. Use the condition $b^2 < 4ac$ to ensure the function never reaches zero.

Problem 2

Evaluate the following limit:

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^5 + y^5}{x^3 + y^3}$$

Hint 1

Start by factoring the numerator using the sum of fifth powers formula and the denominator using the sum of cubes formula. Look for a common algebraic factor, $(x + y)$, that can be canceled out to simplify the expression.

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Hint 2

After simplifying, convert the remaining fractional expression into polar coordinates by letting $x = r \cos \theta$ and $y = r \sin \theta$.

Problem 3

A set K in $\mathbb{R}^m$ is called a convex set if for any $\mathbf{x}_1, \mathbf{x}_2 \in K$ and any $0 < t < 1$, we have $(1 - t)\mathbf{x}_1 + t\mathbf{x}_2 \in K$. A function $f : K \rightarrow \mathbb{R}$ is called a convex function if for any $\mathbf{x}_1, \mathbf{x}_2 \in K$ and any $0 < t < 1$, we have

$$f((1 - t)\mathbf{x}_1 + t\mathbf{x}_2) \leq (1 - t)f(\mathbf{x}_1) + tf(\mathbf{x}_2).$$

Prove that any convex function on an open convex set K is locally Lipschitz, that is, for any $\mathbf{x}_0 \in K$, there exists a neighborhood U of $\mathbf{x}_0$ and a constant $L > 0$ such that for any $\mathbf{x}, \mathbf{y} \in U$, we have

$$|f(\mathbf{x}) - f(\mathbf{y})| \leq L\|\mathbf{x} - \mathbf{y}\|.$$

Hint 1

Construct a polytope around $\mathbf{x}_0$ within K to find a bound M via vertex combinations.

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Hint 2

For any $\mathbf{x}, \mathbf{y} \in B(\mathbf{x}_0, r)$ with distance d , extend the ray from $\mathbf{y}$ through $\mathbf{x}$ to a point $\mathbf{z}$ at distance r from $\mathbf{x}$. Verify $\mathbf{z} \in B(\mathbf{x}_0, 2r)$.

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Hint 3

Express $\mathbf{x}$ as a convex combination: $\mathbf{x} = \frac{r}{r+d}\mathbf{y} + \frac{d}{r+d}\mathbf{z}$. Apply the convexity condition, subtract $f(\mathbf{y})$, and use bound M to derive the Lipschitz constant $L = \frac{2M}{r}$.

Differential of Multivariable Functions

Problem 1

Let $f(x, y)$ be a differentiable function on $\mathbb{R}^2 \setminus \{(0, 0)\}$ and k be a non-negative integer. Suppose that for any $(x, y) \neq (0, 0)$, the following equation holds:

$$x \frac{\partial f}{\partial x}(x, y) + y \frac{\partial f}{\partial y}(x, y) = k f(x, y)$$

Prove that f is a homogeneous function of degree k . That is, for any $(x, y) \neq (0, 0)$ and any positive real number t , show that:

$$f(tx, ty) = t^k f(x, y)$$

Hint 1

Fix an arbitrary point $(x, y) \neq (0, 0)$. To connect $f(tx, ty)$ and $t^k f(x, y)$, construct an auxiliary function of a single variable $t > 0$:

$$g(t) = t^{-k} f(tx, ty)$$

Your ultimate goal is to prove that $g(t)$ is a constant function independent of t .

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Hint 2

Calculate the derivative $g'(t)$ using the product rule and the multivariate chain rule. Then, evaluate the given condition (the partial differential equation) at the scaled point $X = tx$ and $Y = ty$. Substitute this relationship into your expression for $g'(t)$ to show that $g'(t) = 0$. Finally, evaluate $g(1)$ to finish the proof.